MAST20026 REAL ANALYSIS EXAM CHEAT SHEET & REFERENCE MANUAL

Part I: Visual Topic Tree Index | Part II: Axioms, Proof Templates, Counterexamples & Convergence Matrices

IN-SCOPE (Core 9-Question Exam Blueprint)	OUT-OF-SCOPE (Never Tested / Avoid Wasting Time)
• Q1: Truth Tables, Logical Deduction & Equivalent Connective • Q2: Step-by-step Formal Proof from Field/Order Axioms of R • Q3: Upper Bounds & Supremum Proofs (sup(A+B), sup(kA)) • Q4: Constructing Specific Counterexamples (Limit points, Sequences) • Q5: Recursive Sequence Induction, Monotonicity, MCT & Cauchy • Q6: Formal eps-M / eps-delta Limit & Discontinuity Negation Proof • Q7: Riemann Sums L(f,P), U(f,P) & Q / (R\Q) Non-Integrability • Q8: Infinite Series Tests (Comparison, AST, Ratio) & Power Series R • Q9: Taylor Polynomial Lagrange Remainder & Uniform Convergence	• Metric Spaces, General Topological Spaces or Open Ball Topologies • Lebesgue Measure, Lebesgue Integration or Fourier Series • Multivariable Calculus, Partial Derivatives or Multiple Integrals • Complex Analysis or Contour Integration • Memorizing Axiom Numbers (Axioms A1-M5, O1-OM are printed on Q2!) • Using eps-N proofs inside Q8 Series questions (explicitly exempted)

PART I · VISUAL TOPIC TREE INDEX

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[!!!] = Core / Must-know    [!! ] = Important / Frequent    [!  ] = Basic / Edge case

[0.0] MAST20026 REAL ANALYSIS — COMPLETE EXAM SYLLABUS
│
├── [1.0] Logic, Propositional Calculus & Axioms of Real Numbers
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│   │   ├── [1.1.2] 4-Row Truth Table construction & row deduction       [!!!]
│   │   ├── [1.1.3] Equivalent single-connective statement matching      [!!!]
│   │   └── [1.1.4] Quantifier negation (forall <-> exists) & contrapos. [!! ]
│   │
│   └── [1.2] Field & Order Axioms of R (Always Q2!)
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│       ├── [1.2.2] Order Axioms (O1-O2, OA, OM) & Completeness (C)      [!!!]
│       └── [1.2.3] Line-by-line formal axiom cancellation/zero proofs   [!!!]
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├── [2.0] Completeness, Supremum / Infimum & Topology of R
│   ├── [2.1] Upper/Lower Bounds, Supremum & Infimum (Always Q3!)
│   │   ├── [2.1.1] Upper bound definition & Least Upper Bound (sup)     [!!!]
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│   │   └── [2.1.3] Proofs of sup(A+B) = sup A + sup B & sup(kA)=k sup A [!!!]
│   │
│   └── [2.2] Limit Points (Accumulation Points) & Density
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│       ├── [2.2.2] Constructing sets with specific limit point sets     [!!!]
│       └── [2.2.3] Archimedean Property & Density of Q and R\Q in R     [!!!]
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│   ├── [3.1] Formal Sequence Convergence & Limit Laws
│   │   ├── [3.1.1] Formal eps-N limit definition & proof template       [!!!]
│   │   ├── [3.1.2] Squeeze Theorem & algebra of limits                  [!! ]
│   │   └── [3.1.3] Divergent sequence counterexample construction (Q4)  [!!!]
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│   │   └── [4.1.3] Sequential characterization of continuity            [!! ]
│   │
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│   │   ├── [4.2.1] Intermediate Value Theorem (IVT) root existence      [!! ]
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│   └── [4.3] Differentiability, Mean Value Theorem & Taylor's Theorem
│       ├── [4.3.1] First-principles derivative & Rolle's / MVT          [!! ]
│       └── [4.3.2] Taylor polynomial T_k(x) & Lagrange Remainder bound  [!!!]
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│   │
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│       ├── [5.2.2] Bounding Upper Integral > Lower Integral (Non-integ.)[!!!]
│       └── [5.2.3] Fundamental Theorem of Calculus (FTC Parts 1 & 2)    [!! ]
│
└── [6.0] Infinite Series, Power Series & Sequences of Functions (Q8 & Q9!)
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    │   ├── [6.1.1] Divergence Test, Geometric Series & p-Series Test    [!!!]
    │   ├── [6.1.2] Direct Comparison Test & Ratio Test                  [!!!]
    │   └── [6.1.3] Alternating Series Test (AST) & Absolute Convergence [!!!]
    │
    ├── [6.2] Power Series & Radius/Interval of Convergence (Always Q8b!)
    │   ├── [6.2.1] Ratio formula for Radius of Convergence R            [!!!]
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    │
    └── [6.3] Pointwise vs Uniform Convergence of Function Sequences (Q9b!)
        ├── [6.3.1] Pointwise limit f(x) = lim_{n->inf} f_n(x)           [!!!]
        ├── [6.3.2] Supremum norm M_n = sup |f_n(x) - f(x)| -> 0 proof   [!!!]
        └── [6.3.3] Preservation of continuity/integrals & Weierstrass M [!! ]
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PART II · REFERENCE TABLES, AXIOMS & STEP-BY-STEP PROOF TEMPLATES

[1.0] Logic, Truth Tables (Q1) & Axioms of the Real Numbers (Q2)

Index & Topic	Core Rule / Axiom Table	Exam Scoring Requirement & Pitfalls
[1.1.1] Connectives & Implication Rule	• ~p (NOT) | p ^ q (AND: true iff both T) | p v q (OR: false iff both F) • p => q (IMPLIES): FALSE ONLY WHEN p = T and q = F! Whenever p = F, (p => q) is automatically TRUE (Vacuous Truth)!	Standard 4 rows for (p, q): (T,T), (T,F), (F,T), (F,F). Logical Equivalence: p => q <=> (~p) v q <=> (~q) => (~p) Negation of (p => q) is: p ^ (~q)
[1.1.2] Q1(b) & Q1(c) Justification Rules	• Q1(b): Cross-check rows where given premises are T; state exact line numbers (e.g. 'Only Line 2 and Line 4 have q = F, and Line 2 has compound = F, so by Line 4, p must be F').	• Q1(c): Compare final truth column against standard 1-connective columns: p => q (T,F,T,T), q => p (T,T,F,T), p ^ q (T,F,F,F), p v q (T,T,T,F), ~p, ~q.
[1.2.1] Field Axioms (Printed on Exam Q2)	• A1/M1 (Closure): x+y in R, xy in R • A2/M2 (Commutative): x+y = y+x | xy = yx • A3/M3 (Associative): (x+y)+z = x+(y+z) | (xy)z = x(yz)	• A4/M4 (Identity): x + 0 = x | 1 * x = x • A5/M5 (Inverse): exists (-x) s.t. x+(-x)=0 | for x!=0, exists 1/x s.t. x*(1/x)=1 • D (Distributive): x(y + z) = xy + xz
[1.2.2] Order Axioms	• O1 (Trichotomy): exactly one of x < y, y < x, x = y holds • O2 (Transitivity): x < y and y < z => x < z	• OA (Order-Add): x < y => x + z < y + z • OM (Order-Mult): x > 0 and y > 0 => xy > 0

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/* [1.2.3] Q2 Formal Axiom Proof Template 1: Cancellation Law (Assume a + b = a + c, prove b = c) */
1. Since a in R, by Axiom A5 there exists (-a) in R such that a + (-a) = 0.
2. Since a + b = a + c, adding (-a) to both sides gives: (-a) + (a + b) = (-a) + (a + c).   [Substitution]
3. By Axiom A3 (Associativity):   ((-a) + a) + b = ((-a) + a) + c.
4. By Axiom A2 (Commutativity):   (a + (-a)) + b = (a + (-a)) + c.
5. By Step 1 (Axiom A5):          0 + b = 0 + c.
6. By Axiom A2 and Axiom A4 (Additive Identity x + 0 = x):  b = c.  [Q.E.D.]
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[2.0] Completeness, Supremum Proofs (Q3) & Counterexample Bank (Q4)

Index & Concept	Formal Definition / Characterization	Exam Proof Strategy
[2.1.1] Upper Bound & Supremum (sup S)	• alpha is an Upper Bound of S <=> for all x in S, x <= alpha. • alpha = sup S (Least Upper Bound) iff: (1) alpha is an upper bound of S, AND (2) For any upper bound M of S, alpha <= M.	• Golden Trick to prove sup S <= M: Simply prove that M is an upper bound of S! Then since sup S is the LEAST upper bound, sup S <= M follows immediately!
[2.1.2] Analytic eps-Characterization	Assume alpha is an upper bound of S. Then: alpha = sup S <=> for all eps > 0, there exists x in S such that x > alpha - eps.	Use this when proving sup(kA) = k sup A (with k > 0): let eps > 0, since sup A - eps/k is not an upper bound of A, pick x in A with x > sup A - eps/k, multiply by k!
[2.2.1] Limit Point (Accumulation Point)	L in R is a limit point of subset E subset of R iff: For every eps > 0, ((L - eps, L + eps) \ {L}) intersect E != empty. (Equivalently: there exists a sequence x_n in E \ {L} with lim x_n = L).	• Finite sets have NO limit points (limit point set = empty). • Limit point of {1/n : n in N+} is {0}. • Limit point of {2 + 1/n : n in N+} is the single-element set {2}.

[2.2.2 / 3.1.3] Q4 High-Frequency Exam Prompt ('Provide a specific example...')	Ready-to-Write Concrete Example & 1-Line Justification
1. Infinite subset E subset of R whose set of limit points is {c} (e.g. {2})	Let E = { 2 + 1/n : n in N+ }. Since lim_{n->inf} (2 + 1/n) = 2 with terms distinct from 2, 2 is the unique limit point.
2. Infinite subset E subset of R whose set of limit points is {a, b} (e.g. {0, 1})	Let E = { 1/n : n in N+ } union { 1 + 1/n : n in N+ }. The two subsequences accumulate at 0 and 1 respectively.
3. Sequence (a_n) where (a_n)^2 converges to K (e.g. 16), but (a_n) diverges	Let a_n = 4 * (-1)^n. Then (a_n)^2 = 16 -> 16 (converges), while a_n alternates between -4 and 4 (diverges).
4. Two divergent sequences (x_n), (y_n) whose sum (x_n + y_n) converges to L (e.g. 3)	Let x_n = 3 + (-1)^n and y_n = -(-1)^n. Both oscillate and diverge, but x_n + y_n = 3 -> 3 (converges).
5. Two divergent sequences (x_n), (y_n) whose product (x_n * y_n) converges to 1	Let x_n = (-1)^n and y_n = (-1)^n. Both diverge, but x_n * y_n = ((-1)^n)^2 = 1 -> 1.
6. Bounded sequence that diverges (not Cauchy) but has convergent subsequences	Let x_n = (-1)^n. Bounded in [-1, 1], diverges; even subsequence x_{2k} = 1 -> 1, odd x_{2k-1} = -1 -> -1.
7. Decreasing sequence of positive terms a_n -> 0 whose series sum a_n DIVERGES	Let a_n = 1/n (Harmonic series) or a_n = 1/sqrt(n). Here a_n -> 0, but sum_{n=1}^inf (1/n) diverges (p = 1 <= 1).

[3.0] Sequences: Formal eps-N Proofs & Recursive MCT / Cauchy (Q5)

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/* [3.1.1] Formal eps-N Sequence Limit Proof Template: Prove lim_{n->inf} (2n + 1)/(n + 3) = 2 */
[Scratch Work]: |(2n+1)/(n+3) - 2| = |(2n+1 - 2n - 6)/(n+3)| = 5/(n+3) < 5/n. Want 5/n < eps <=> n > 5/eps.
[Formal Proof]: Let eps > 0 be given. Choose N = ceil(5 / eps) (or by Archimedean property, pick N in N+ with N > 5/eps).
                Then for all n > N, we have:
                    |(2n + 1)/(n + 3) - 2| = |-5| / (n + 3) = 5 / (n + 3) < 5 / n < 5 / N <= 5 / (5 / eps) = eps.
                Therefore, by definition of the limit, lim_{n->inf} (2n + 1)/(n + 3) = 2.  [Q.E.D.]
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[3.2] Q5 Recursive Sequence 3-Part Blueprint (13 Marks)	Exact Step-by-Step Write-Up & Formula
Part (a): Boundedness by Induction [5 marks] Prove a <= x_n < b for all n in N+ (e.g. x_1 = 0, x_{n+1} = 9/(6 - x_n), prove 0 <= x_n < 3)	1. Base Case (n = 1): State x_1 = 0, which clearly satisfies 0 <= 0 < 3. Thus true for n = 1. 2. Inductive Hypothesis: Suppose 0 <= x_k < 3 holds for some k in N+. 3. Inductive Step (n = k + 1): Chain inequalities from IH: 0 <= x_k < 3 => -3 < -x_k <= 0 => 3 < 6 - x_k <= 6 => 1/6 <= 1/(6 - x_k) < 1/3 => 9/6 <= 9/(6 - x_k) < 9/3 => 0 <= 1.5 <= x_{k+1} < 3. 4. Mandatory Closing Sentence: 'By the Principle of Mathematical Induction, 0 <= x_n < 3 holds for all n in N+.'
Part (b): Monotonicity [4-5 marks] Compute x_{n+1} - x_n and determine sign	1. Combine over common denominator: x_{n+1} - x_n = 9/(6 - x_n) - x_n = (9 - 6x_n + x_n^2)/(6 - x_n) = (x_n - 3)^2 / (6 - x_n). 2. Analyze signs using Part (a) bound (0 <= x_n < 3): Since x_n != 3, numerator (x_n - 3)^2 > 0; denominator 6 - x_n > 3 > 0. 3. Conclusion: Since x_{n+1} - x_n > 0 for all n in N+, (x_n) is strictly monotone increasing!
Part (c): Cauchy & Limit L [3-4 marks] Is (x_n) Cauchy? Find lim x_n	1. MCT: Since (x_n) is monotone increasing and bounded above by 3, by the Monotone Convergence Theorem (MCT), (x_n) converges to some limit L in R. 2. Cauchy: Since every convergent sequence in R is a Cauchy sequence, YES, (x_n) is Cauchy. 3. Fixed-Point Limit: Taking n -> inf on both sides of x_{n+1} = 9/(6 - x_n) gives L = 9/(6 - L) => L^2 - 6L + 9 = 0 => (L - 3)^2 = 0 => L = 3.

[4.0] Function Limits, Discontinuity (Q6) & Taylor's Theorem (Q9a)

Index & Proof Type	Formal Quantifier Definition	Exam Template & Selection Rule
[4.1.1] Limit as x -> inf (eps-M Proof, Q6a)	lim_{x->inf} f(x) = L <=> for all eps > 0, there exists M in R such that for all x > M, |f(x) - L| < eps.	For piecewise f(x) = 2x/(x+1) when x >= 1 (prove lim = 2): • Simplify |2x/(x+1) - 2| = 2/(x+1) < 2/x < eps <=> x > 2/eps. • Choose M = max{1, 2/eps} (ensures x > 1 branch AND 2/x < eps)!
[4.1.1] Limit as x -> a (eps-delta Proof)	lim_{x->a} f(x) = L <=> for all eps > 0, there exists delta > 0 s.t. 0 < |x - a| < delta => |f(x) - L| < eps.	For nonlinear polynomial (e.g. x^2 near a = 3): factor |x^2 - 9| = |x - 3||x + 3|. Restrict delta <= 1 first so |x - 3| < 1 => 2 < x < 4 => |x + 3| < 7. Then choose delta = min{1, eps / 7}!
[4.1.2] Discontinuity at x = a (Negation Proof, Q6b)	f is NOT continuous at x = a <=> there exists eps_0 > 0 such that for all delta > 0, there exists x in R with |x - a| < delta AND |f(x) - f(a)| >= eps_0.	For piecewise jump at a = 1 (e.g. f(x) = 3x - 5 for x < 1, and f(1) = 1): 1. Left limit is 3(1)-5 = -2; f(1) = 1. Jump gap = |1 - (-2)| = 3. 2. Pick concrete eps_0 = 1 (or 1/2). Let arbitrary delta > 0 be given. 3. Pick x = 1 - min{delta/2, 1/2} < 1. Then |x - 1| < delta, and f(x) = 3x - 5 < -2, so |f(x) - f(1)| = 1 - (3x - 5) = 6 - 3x > 3 > eps_0 = 1!
[4.3.2] Taylor's Theorem & Lagrange Remainder (Q9a)	T_k(x) = sum_{m=0}^k [f^(m)(a) / m!] (x - a)^m Remainder: R_k(x) = f(x) - T_k(x) = [f^(k+1)(c) / (k+1)!] (x - a)^(k+1) for some c strictly between a and x.	Example (g(x) = cos(x) - 1 at a = 0, degree k = 3): • g(0)=0, g'(0)=0, g''(0)=-1, g'''(0)=0 => T_3(x) = -x^2 / 2. • (k+1)=4th derivative: g^(4)(c) = cos(c). Since |cos(c)| <= 1 for all c in R, |g(x) - T_3(x)| = |cos(c) / 4!| |x|^4 <= |x|^4 / 24 = x^4 / 24.

[5.0] Riemann Integration, Darboux Sums & Dirichlet Non-Integrability (Q7)

[5.1 / 5.2] Q7 3-Part Exam Blueprint (13 Marks)	Formulas, Density Citation & Full-Mark Justification
Part (a): Concrete Partition Calculation [4 marks] Given g(x) on [a, b] and explicit partition P = {x_0, x_1, ..., x_n}, compute L(g, P) and U(g, P)	• Subinterval i is I_i = [x_{i-1}, x_i] with width Delta x_i = x_i - x_{i-1}. • m_i = inf_{x in I_i} g(x) and M_i = sup_{x in I_i} g(x). (If g is increasing on I_i: m_i = g(x_{i-1}), M_i = g(x_i). If decreasing: m_i = g(x_i), M_i = g(x_{i-1})). • Lower Sum: L(g, P) = sum_{i=1}^n m_i * Delta x_i | Upper Sum: U(g, P) = sum_{i=1}^n M_i * Delta x_i.
Part (b): Arbitrary Partition on Q vs R\Q Function [4 marks] Let f(x) = { h(x) >= 0 if x in Q ; 0 if x not in Q } on [0, 2]. Compute L(f, P) for ANY partition P = {x_0, ..., x_n}	1. Cite Density of Irrationals: 'By the density of the irrationals (R \ Q) in R, every subinterval [x_{i-1}, x_i] contains at least one irrational number x, where f(x) = 0.' 2. Since f(x) >= 0 everywhere on [0, 2], we have m_i = inf_{x in [x_{i-1}, x_i]} f(x) = 0 for every i = 1, ..., n. 3. Therefore, L(f, P) = sum_{i=1}^n 0 * (x_i - x_{i-1}) = 0. Consequently, the Lower Riemann Integral is sup_P L(f, P) = 0!
Part (c): Positive Lower Bound on Upper Integral & Non-Integrability [5 marks] For f(x) = { 4 - x^2 if x in Q ; 0 if x not in Q } on [0, 2], bound Upper Integral > 0 and prove f is NOT integrable	1. Cite Density of Rationals: By the density of Q in R, on any subinterval inside [0, 1], M_i = sup f(x) >= inf_{x in [0,1] intersect Q} (4 - x^2) = 4 - 1^2 = 3. (Or simpler: for any partition P, let P' = P union {1}. On [0, 1], since 4 - x^2 >= 3 on Q, every subinterval of P' in [0, 1] has M_i >= 3, so U(f, P) >= U(f, P') >= 3 * (1 - 0) = 3 > 0). 2. Thus the Upper Riemann Integral satisfies inf_P U(f, P) >= 3 > 0. 3. Since Upper Integral (>= 3) != Lower Integral (= 0), f is NOT Riemann integrable on [0, 2]!

[6.0] Infinite Series, Power Series (Q8) & Uniform Convergence (Q9b)

[6.1] Series Test	Formula / Hypotheses to Check	Conclusion & Exam Application
1. Divergence Test	Compute lim_{n->inf} a_n	If lim a_n != 0 (or DNE), series sum a_n DIVERGES! (If lim = 0, test is inconclusive)
2. Geometric Series	sum_{n=0}^inf a * r^n	Converges to a / (1 - r) iff |r| < 1; Diverges if |r| >= 1
3. p-Series Test	sum_{n=1}^inf (1 / n^p)	CONVERGES iff p > 1 (e.g. 1/n^2, 1/n^3); DIVERGES iff p <= 1 (e.g. 1/n, 1/sqrt(n))
4. Direct Comparison Test (a_n >= 0)	• To prove Convergence: find 0 <= a_n <= b_n where sum b_n converges • To prove Divergence: find 0 <= b_n <= a_n where sum b_n diverges	Example (2025 Q8): 4 / (n^3 + 5). Since n^3 + 5 > n^3 > 0 for all n >= 1, 0 < 4 / (n^3 + 5) < 4 / n^3. Since sum 4(1/n^3) converges (p = 3 > 1), series converges!
5. Alternating Series Test (AST)	For sum (-1)^n b_n with b_n > 0, check 3 conditions: (1) b_n > 0 | (2) (b_n) is decreasing (b_{n+1} <= b_n) | (3) lim_{n->inf} b_n = 0	Example: sum (-1)^n / sqrt(n + 3). Here b_n = 1/sqrt(n+3) > 0, strictly decreasing since sqrt(n+4) > sqrt(n+3), and lim b_n = 0 => CONVERGES by AST!
6. Ratio Test	Compute L = lim_{n->inf} |a_{n+1} / a_n|	• L < 1 => Converges absolutely • L > 1 (or inf) => Diverges • L = 1 => Inconclusive Use whenever a_n contains factorials (n!) or exponentials (c^n)!
[6.2] Power Series sum c_n (x - a)^n	1. Radius R = lim_{n->inf} |c_n / c_{n+1}| (or apply Ratio Test |(x-a)| * L < 1) 2. Test left endpoint x = a - R and right endpoint x = a + R separately!	Example: sum (2^n / sqrt(n)) x^n. R = lim |(2^n/sqrt(n)) / (2^{n+1}/sqrt(n+1))| = 1/2. • At x = -1/2: sum (-1)^n / sqrt(n) -> Converges by AST! • At x = +1/2: sum 1 / sqrt(n) -> Diverges (p = 1/2 <= 1)! Interval I = [-1/2, 1/2).

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/* [6.3.2] Q9(b) Uniform Convergence Supremum Proof Template: f_n(x) = x^2 / (n + x^2) on [0, 1] */
1. Pointwise Limit: Fix x in [0, 1]. Since 0 <= x^2 <= 1, as n -> inf, lim_{n->inf} x^2 / (n + x^2) = 0.
                    Thus the pointwise limit function is f(x) = 0 for all x in [0, 1].
2. Uniform Convergence (M_n Bound): For all x in [0, 1], since x^2 <= 1 and n + x^2 >= n > 0, we have:
                    |f_n(x) - f(x)| = |x^2 / (n + x^2) - 0| = x^2 / (n + x^2) <= 1 / n.
   Therefore, M_n = sup_{x in [0,1]} |f_n(x) - f(x)| <= 1 / n.
   Since lim_{n->inf} (1 / n) = 0, by Squeeze Theorem lim_{n->inf} M_n = 0. Hence f_n -> f uniformly on [0, 1]. [Q.E.D.]
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